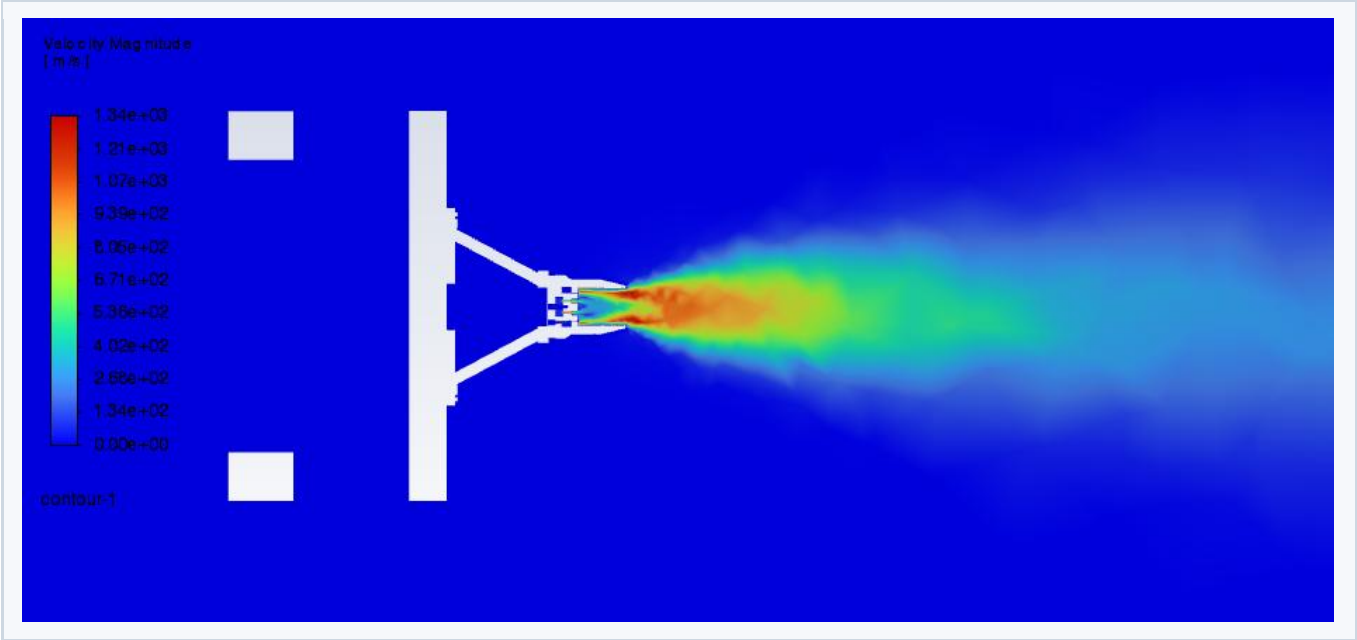
KOTA YASASVI SURYA VIGNESH

Aerospace Engineering Portfolio

University of Petroleum and Energy Studies (UPES), Dehradun

Degree: B.Tech in Aerospace Engineering (2023–2027)
Primary Disciplines: Rocket Propulsion, Reactive Flows, High-Speed CFD, Aerostructures, Space Mission Design & STK, and UAV Systems

Phone: +91 7075500780
Email: ksuryavigneshkysv3252@gmail.com
LinkedIn: https://linkedin.com/in/surya-vignesh-kota



SPECIALIZED TECHNICAL DISCIPLINES

- **Propulsion, Spray Mechanics & Reactive Aerothermodynamics:** Liquid rocket pintle injector atomization modelling (discrete phase particle tracking & empirical formulations), RDE and Ramjet combustor geometry generation, torch ignitors, and altitude-compensating annular aerospike nozzles.
- **High-Speed Computational Fluid Dynamics:** Pressure- and density-based coupled solvers in ANSYS Fluent, multi-regime compressible flow modelling, 2D planar/axisymmetric combustion setups, supersonic shock pattern evaluation, and inviscid Method of Characteristics (MOC) contour implementation.
- **Aircraft Aerodynamics & Mission Flight Mechanics:** Airfoil aerodynamic polar characterization using XFLR5, canard-wing vortex coupling analysis, drag polar synthesis (C_{D0}, induced drag factor K), power-required estimation, and operational flight envelope (V-n diagram) construction.
- **Aerostructures & Finite Element Analysis (FEA):** Internal structural framework sizing (spars, transverse ribs, skin stringers) in CATIA V5 and SolidWorks; coupled Multiphysics fluent-structural setups via ANSYS System Coupling; fixed-cantilever and inertia-relief static structural FEA in ANSYS Mechanical.
- **Unmanned Aerial Systems (UAV) Integration:** Multirotor airframe structural modelling (F450 center plates, arm brackets), 2-axis stabilized gimbal design with vibration isolation, and payload packaging for automated airborne thermal inspection missions.
- **Space Mission Design & Orbital Mechanics:** Low Earth Orbit (LEO) constellation trade studies in Ansys STK; parametric coverage modeling (Figures of Merit), nodal regression shift analysis, optical sensor field-of-view (FOV) footprint sizing, and ground-station/regional target access window optimization.

COMPUTATIONAL ENGINEERING STACK

CAD & Generative Design
CATIA V5, SolidWorks, SpaceClaim, AutoCAD.

CFD & Fluid Solvers
ANSYS Fluent (Pressure/Density-based, DPM, Species Transport), XFLR5, OpenRocket, Ansys STK (Systems Tool Kit).

Structural & Multiphysics FEA:
ANSYS Mechanical (Static Structural, Inertia Relief), ANSYS System Coupling, COMSOL Multiphysics.

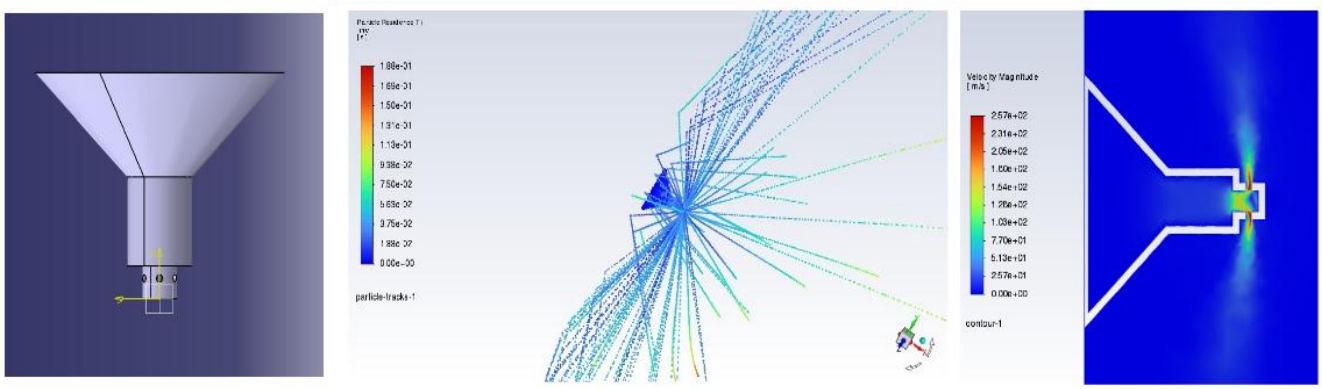
Scientific Computing & Code
MATLAB (MOC Solvers, Multi-variable Parametric Matrices), Python (Data Analysis, Numerical Methods).

ROCKET PROPULSION & MULTIPHASE SPRAY DYNAMICS

Primary Technical Focus: Pintle Injector Primary Breakup, DPM Tracking & Empirical Modeling

Tools: ANSYS Fluent, MATLAB, CATIA V5

Project Reference: *Major Capstone Project — Aug 2026–Present*

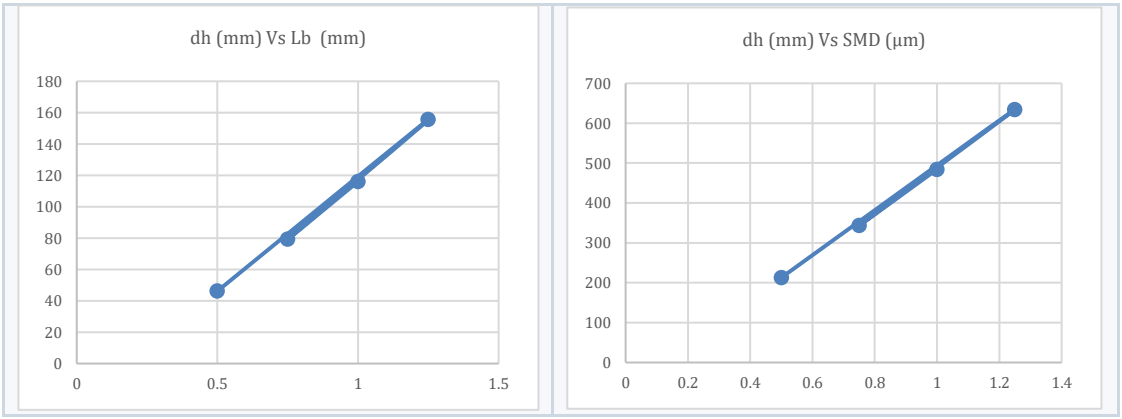


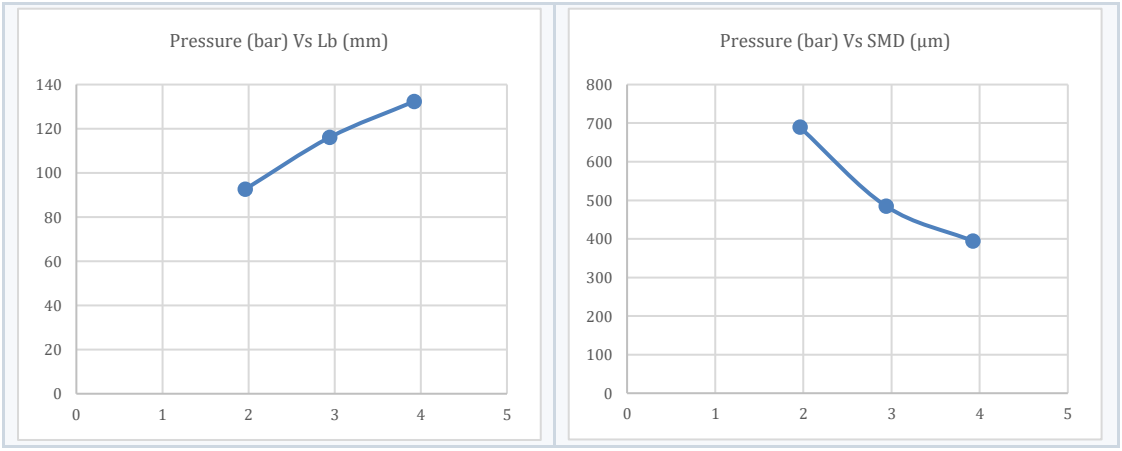
Injector CAD & Computational Tracking

- **Hardware Replication:** Modeled a high-fidelity 3D CAD replica in CATIA V5 matching the liquid propulsion experimental rig at UPES.
- **Configuration:** Single-propellant radial-hole pintle geometry with dedicated internal liquid manifold channels.
- **Multiphase DPM Setup:** Employed ANSYS Fluent pressure-based solver coupled with the Discrete Phase Model (DPM).
- **Particle Tracking:** Injected and tracked 500 discrete sample particles over 1.0 second as a proof-of-concept run to observe qualitative spray cone tendencies within student license mesh constraints.

20-Parameter MATLAB Empirical Framework

- **Parametric Architecture:** Programmed an exhaustive numerical solver taking 20 geometric, fluid, and pressure inputs:
- **Geometric:** Hole diameter, pintle opening distance, spacing.
- **Fluid Properties:** Density, dynamic viscosity, surface tension.
- **Flow States:** Mass flow rate, injection ΔP, momentum ratio.
- **Predictive Analytics:** Computed Sauter Mean Diameter (D_{32}), breakup intact length (L_B), and total spray cone angle (θ).
- **Sensitivity Curves:** Programmed empirical relations in MATLAB to bypass numerical mesh limits, mapping theoretical sensitivity trends of pressure variations on estimated droplet sizes.





REACTIVE FLOWS, TURBOMACHINERY & SYSTEM COUPLING

Primary Technical Focus: 2D Combustion, RDE Test Rig Coupling, Turbomachinery & MOC Nozzles

Experience Reference: *Technical Internship, Prime Toolings — June–July 2026*

Tools: CATIA V5, ANSYS Fluent, System Coupling, MATLAB

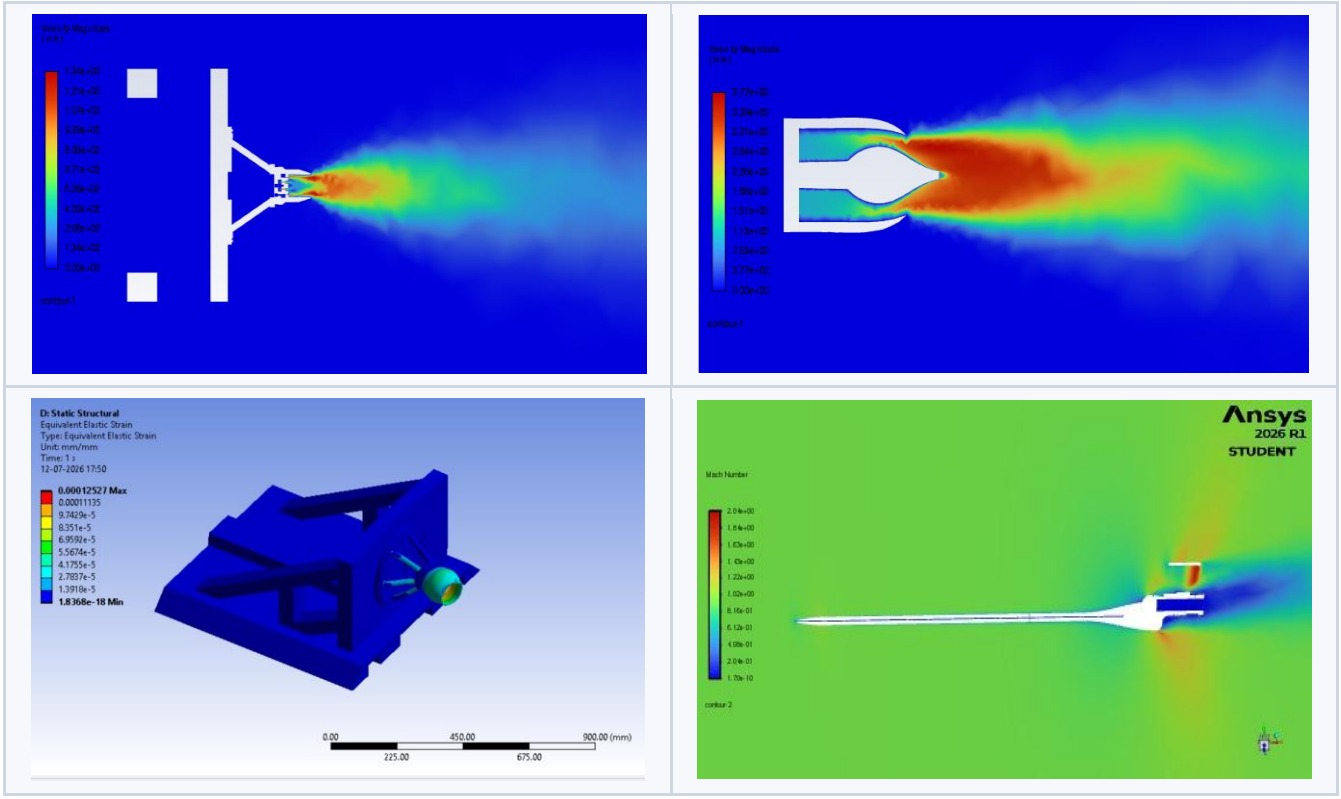


Mechanical CAD & Hardware Modeling

- **Propulsion Hardware:** Modeled high-precision turbine blades, axial stator vanes, and torch ignitor assemblies in CATIA V5.
- **Feed Components:** Developed annular aerospike nozzles and pintle injectors with integrated mounting hardware.
- **Sensor Integration:** Modelled Pitot-static and internal flow-through airspeed sensor geometries for experimental testing.
- **Method of Characteristics (MOC):** Utilized MOC-generated coordinates to model and evaluate inviscid supersonic nozzle contours for fabrication.

ANSYS Fluent & Multiphysics System Coupling

- **2D Combustion Simulation:** Conducted planar and axisymmetric reactive flow simulations evaluating species reaction and temperature rise.
- **Aerospike Combustion:** Resolved flame stabilization and high-enthalpy exhaust acceleration over the central spike contour.
- **System Coupling Execution:** Successfully coupled ANSYS Fluent with Static Structural to transfer aerodynamic pressure and thermal loads to the RDE test rig.
- **Integrity Verification:** Evaluated mechanical deflection and thermal stresses to ensure test rig safety under detonation firings.

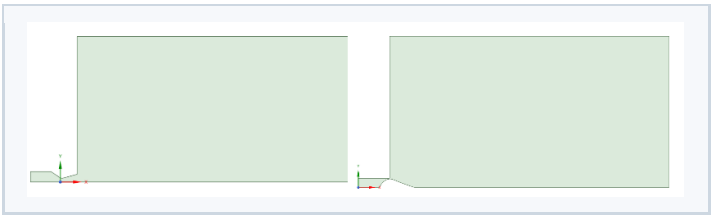


COMPRESSIBLE GAS DYNAMICS & ALTITUDE COMPENSATION

Primary Technical Focus: Parametric SpaceClaim Geometry, Method of Characteristics & CD vs. Spike CFD

Tools: SpaceClaim, ANSYS Fluent, MATLAB

Project Reference: *Advanced CFD Investigation — July 2026–Present*

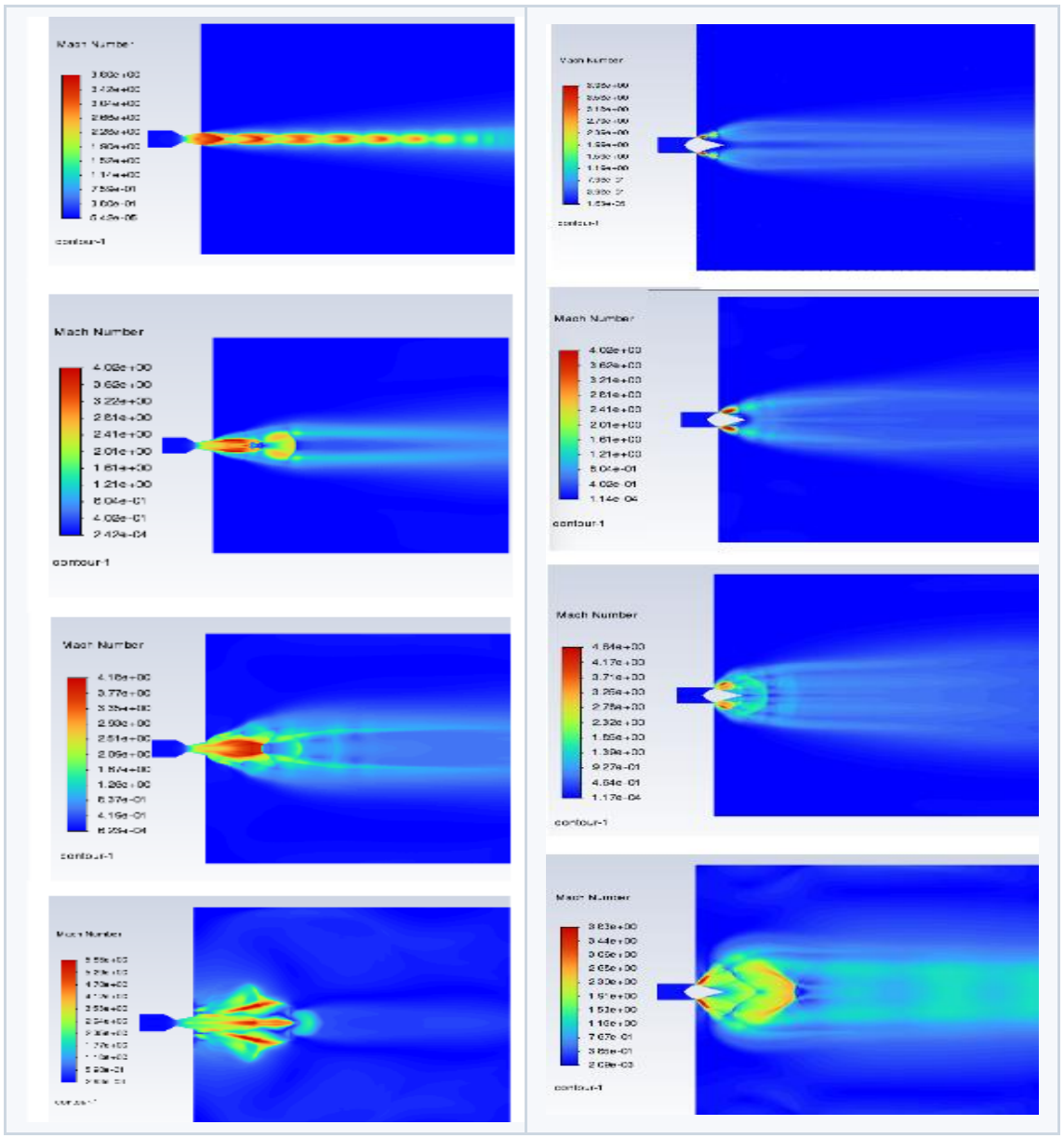


Nozzle Sizing & SpaceClaim Geometry

- **MOC Contour Implementation:** Generated supersonic expansion profiles using Method of Characteristics (MOC) coordinates for both the Bell CD and aerospike nozzles.
- **Matched Throat Area Sizing:** Fixed throat area at $A_t = 314.16\ \text{mm}^2$ across both designs to ensure identical mass flow rates at $P_0 = 2$ MPa and $T_0 = 300$ K.
- **Domain Dimensions:** Modelled the Bell nozzle at $r_t = 10$ mm and the aerospike with a 2.0 mm annular gap ($r_{\text{inner}} = 24$ mm, $r_{\text{outer}} = 26$ mm), aligning the plug tip to $Y = 0$ for 2D axisymmetric solver stability.

4-Altitude Operating Envelopes

- **Atmospheric Test Matrix:** Simulated 4 flight regimes: Sea Level (0 km / 1.013 bar), 5 km (0.540 bar), 10 km (0.265 bar), and 15 km (0.121 bar).
- **Solver Setup:** Solved 2D axisymmetric compressible Navier-Stokes equations in ANSYS Fluent using a density-based coupled solver with k–ω SST turbulence modelling.
- **Altitude Adaptation:** Captured free-boundary plume compensation; aerospike eliminated severe internal shock separation observed in the over-expanded sea-level Bell nozzle.

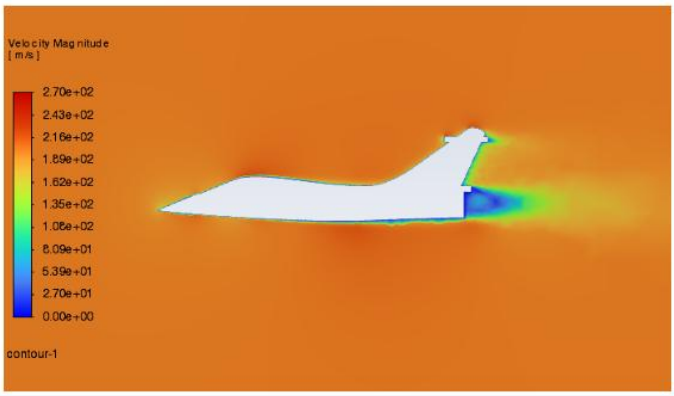


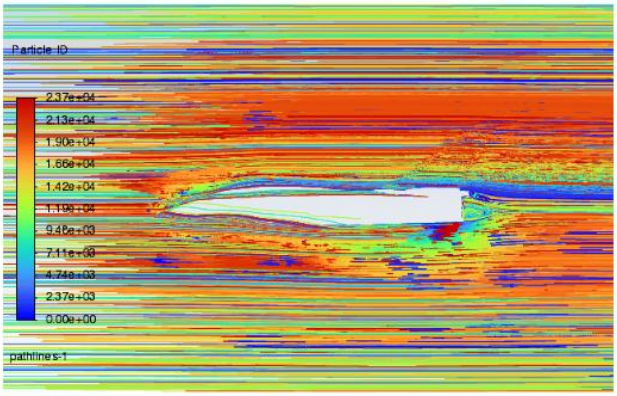
APPLIED FLIGHT DYNAMICS & PERFORMANCE ENVELOPE

Primary Technical Focus: Mission Performance Polar, Rate of Climb, Glide & 3D Viscous CFD

Tools: ANSYS Fluent, MATLAB, Flight Mechanics Formulation

Project Reference: *Dassault Rafale PBL Project — Apr–June 2026*



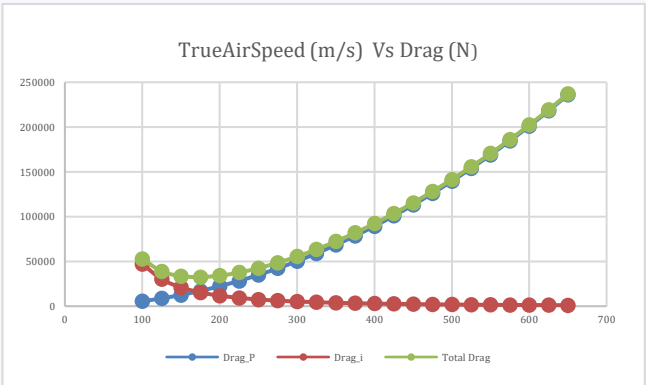


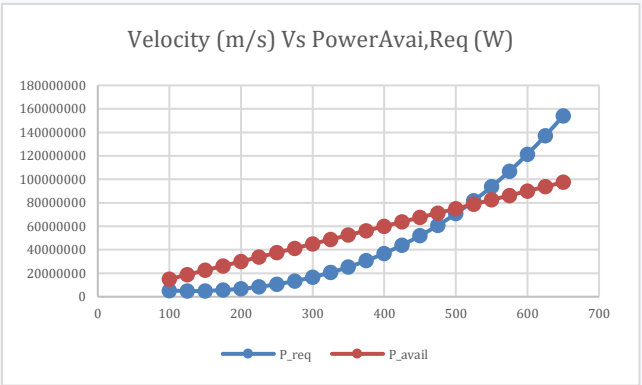
Analytical Mission Performance Sizing

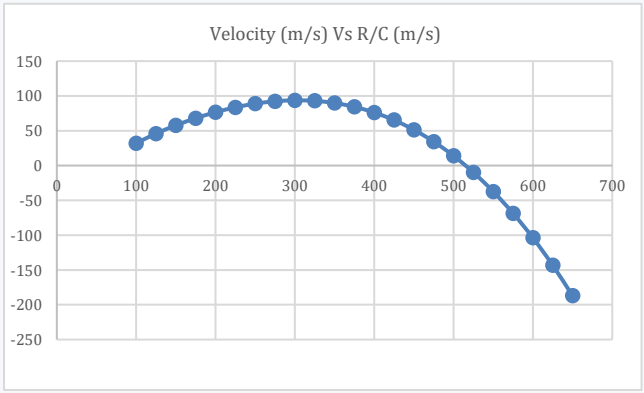
- **Geometric Modeling:** Formulated baseline wing area (S = 45.6 m^2), aspect ratio (AR = 2.60), empty weight, and MTOW parameters.
- **Drag Polar Estimation:** Modeled zero-lift drag coefficient (C_{D0}) and induced drag factors across operational Mach regimes.
- **Flight Envelopes:** Calculated thrust/power required curves, maximum glide ratio (L/D)max, and maximum rate of climb (R/C) across clean and fueled weight conditions.
- **Mission Limits & Glide Profile:** Evaluated maximum aerodynamic efficiency ($(L/D)_{max} \approx 9.30$ at 175 m/s), maximum unpowered glide range (> 102 km from 11 km cruise altitude), and maximum rate of climb ($R/C > 93$ m/s).

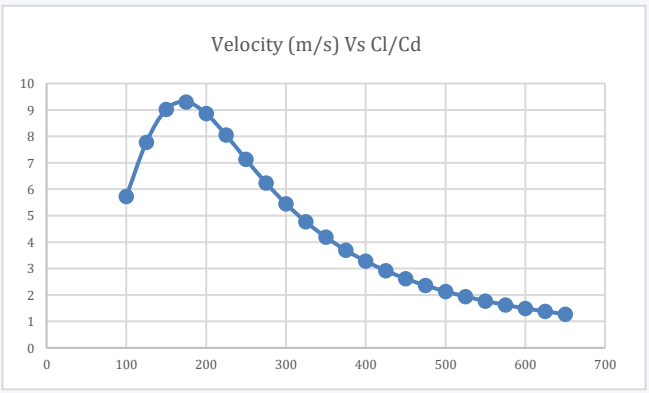
Preliminary 3D Computational Flow Study

- **Airframe Discretization:** Imported an external 3D Rafale delta-canard CAD geometry into ANSYS Fluent using a pressure-based coupled solver.
- **Mesh Resolution:** Constrained to student edition boundaries (512k cells); concentrated cell density along the nose, canard, and wing leading edges without full boundary layer refinement.
- **Qualitative Flow Capture:** Observed qualitative canard vortex shedding and downwash interaction over the delta wing upper surface.
- **Sample Trend Extraction:** Extracted preliminary lift and drag trends as an exploratory numerical study to observe baseline flow behavior alongside analytical estimates.







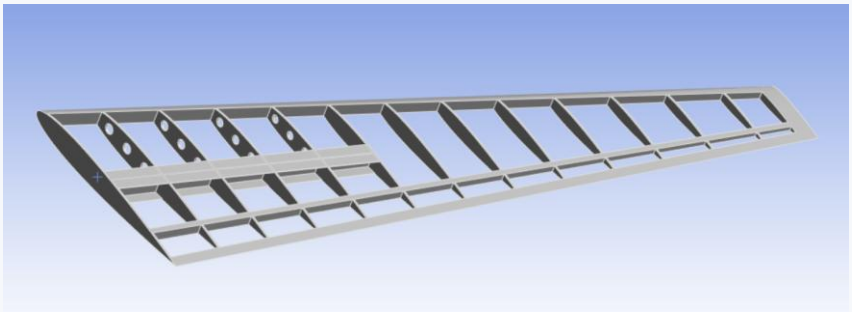


AEROSTRUCTURES & FINITE ELEMENT ANALYSIS

Primary Technical Focus: Tapered Wing Internal Framework, Spar/Rib Architecture & Structural FEA

Tools: ANSYS Mechanical (Static Structural & Modal), CATIA V5, SolidWorks

Project Reference: *Aircraft Structural Design Project — Apr 2026–Present*

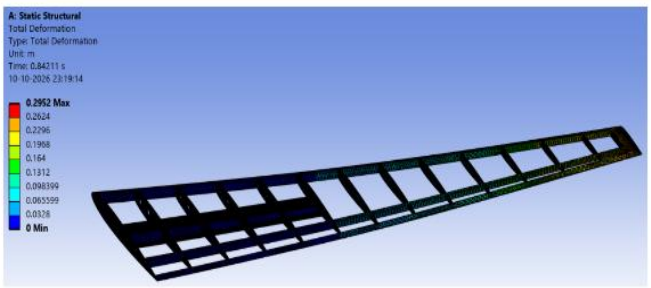


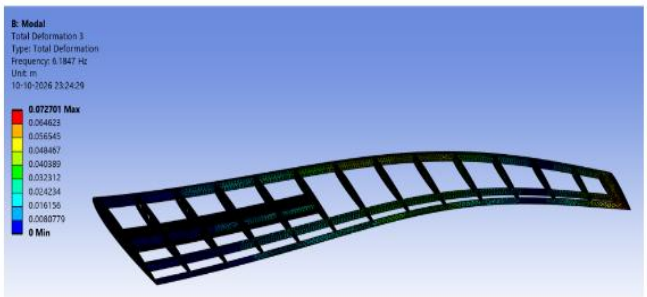
Wing Box Structural Architecture

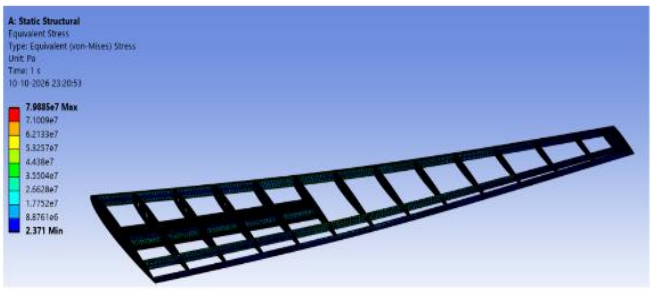
- **Structural Framework:** Discretized a 10.0 m semi-span tapered wing internal framework comprising front and rear I-beam spars, 13 transverse ribs with weight-reduction cutouts, and spanwise stringers.
- **Material Specification:** Modelled using Aerospace Grade Aluminium 7075-T6 ($F_{ty} = 503$ MPa, $E = 71.7$ GPa, $\nu = 0.33$) to evaluate structural strength-to-weight margins.
- **Boundary Conditions:** Enforced cantilever Fixed Support ($U = 0$, $\theta = 0$) across root spar caps, applied a distributed 30 kN vertical load across lower spar caps and rib flanges.

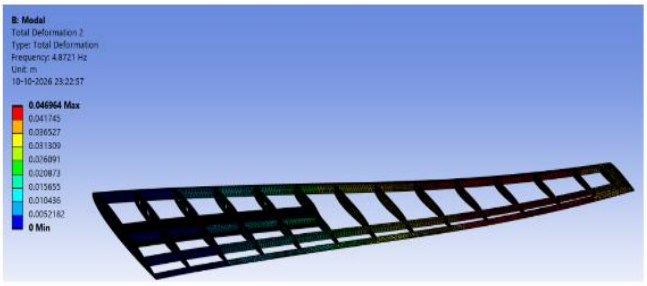
FEA Methodology & Current Results

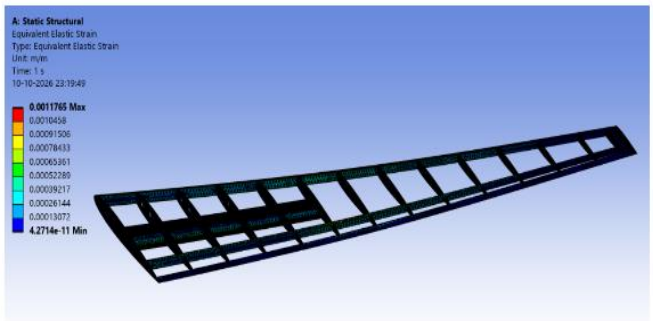
- **Static Deflection & Compliance:** Solved static structural response under 30 kN upward loading, resolving a maximum elastic tip deflection of 0.295 m (2.95% of semi-span) to capture realistic cantilever bending compliance.
- **Stress & Strain Distribution:** Peak equivalent von Mises stress reached 79.89 MPa concentrated at the root spar caps ($< 16\%$ of $F_{ty} = 503$ MPa), yielding an elastic strain of 1.18×10^{-3} m/m and a positive static margin of safety.
- **Dynamic Modal Extraction:** Extracted the first 6 natural frequencies, identifying fundamental out-of-plane bending (Mode 1 at 1.49 Hz), second bending (Mode 2 at 4.87 Hz), and torsion (Mode 3 at 6.18 Hz) to evaluate flutter frequency separation.

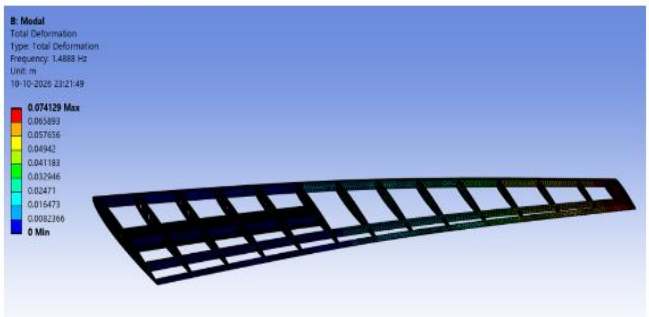












UNMANNED SYSTEMS INTEGRATION & AIRBORNE THERMOGRAPHY

Primary Technical Focus: Airframe Design, Inertia Relief FEA & Thermal Payload Integration

Tools: SolidWorks, CATIA V5, ANSYS Mechanical

Experience Reference: *Hypstuma (2025) & Solar Inspection UAV Project (2026)*



UAV Airframe Design (F450 & Solar Drone)

- **Hypstuma Internship:** Modeled F450 quadcopter airframe components in SolidWorks, focusing on top/bottom center plates and arm joints.
- **Custom Drone Requirements:** Sized a customized multirotor platform for automated solar farm inspections:
- **Total All-Up Weight (AUW):** 3.0 kg.
- **Mission Envelope:** Constant flight profile at 20 m altitude for panel scanning.
- **Thermal Gimbal Packaging:** Designed a 2-axis stabilized radiometric camera mount with anti-vibration damping grommets.

Finite Element Verification & Inertia Relief

- **Fixed-Support FEA:** Analyzed single-arm cantilever bending stress under peak motor thrust to prevent joint failure.
- **Inertia Relief Static Simulation:** Conducted full-vehicle unconstrained equilibrium FEA in ANSYS Mechanical:
- Applied 2.5 kg localized payload mass at vehicle Center of Gravity (CG).
- Balanced by 1.5 kgf upward thrust vectors at each motor base.
- **Structural Margin:** Confirmed stress distributions remained well within yield limits under high-acceleration flight loads.



ORBITAL MECHANICS & SATELLITE CONSTELLATION TRADE STUDY

Primary Technical Focus: LEO Constellation Sizing, 20° Sensor Footprint, Ground Track & Target Revisit Trade Study

Tools: Ansys STK (Systems Tool Kit), MATLAB

Experience Reference: *Space Mission Analysis & Design (SMAD) Investigation (Oct 2026)*

Mission Parameter	500 km LEO Baseline	600 km LEO (Optimal)	700 km LEO Baseline	Design Trend
Sensor Half-Cone Angle	20.0° Nadir	20.0° Nadir	20.0° Nadir	Fixed optical payload
Target Location	Nagpur (Study_Target)	Nagpur (Study_Target)	Nagpur (Study_Target)	Regional target asset
1-Day Coverage (13-14 Oct)	35.95% (165.6M km²)	~41.5% (Interpolated)	46.85% (215.9M km²)	Expands with horizon
1-Day Target Passes	0 Passes	1 Pass (55.47 s)	0 Passes	600 km Day-1 access
7-Day Coverage (13-20 Oct)	52.63% (242.5M km²)	76.76% (353.7M km²)	80.28% (369.9M km²)	Closes nodal gaps
7-Day Target Passes	0 Passes	6 Passes	4 Passes	600 km optimizes revisit
Total Target Contact Time	0.000 s	282.131 s (Max)	260.563 s	600 km maximizes downlink
Mean / Max Pass Duration	N/A	47.02 s / 55.47 s	65.14 s / 79.83 s	Higher arc duration

Orbit Architecture & Mission Sizing

- **Constellation Formulation:** Modeled Low Earth Orbit (LEO) circular architectures comparing 500 km, 600 km, and 700 km altitudes to evaluate Earth observation coverage efficiency.
- **Payload Specification:** Defined a nadir-pointing optical imager configured with a 20.0° half-cone angle.
- **Target & Grid Boundary:** Evaluated regional access over Nagpur (Study Target) and global coverage across −60° to +70° latitude bounds using a 6.0° resolution grid (1,008 assessment points).
- **Temporal Windows:** Executed high-precision orbital propagation across diurnal (24-hr) and multi-day (7-day repeat) cycles to evaluate nodal regression effects.

Coverage FOM & Target Revisit Analysis

- **Global Coverage (FOM):** Demonstrated that increasing altitude from 500 km to 700 km expands 7-day cumulative mapping from 52.63% to 80.28%, closing inter-track blind spots.
- **Nagpur Access Discontinuity:** At 500 km, narrower swath width resulted in zero passes across both 1-day and 7-day intervals due to ground-track shift gaps.
- **Optimal 600 km Regime:** While 700 km yielded maximum total area (80.28%), the 600 km **orbit proved optimal for regional monitoring**, delivering **6 access passes** and 282.13 s **total contact time** (+21.57 s more than 700 km).
- **Mission Trade-off:** Established the balance between wide-area horizon coverage and target revisit density for remote-sensing operations.

